

AHU screening (single-system, 2026-06)

Offline pandas-oracle pack for one AHU folder or a building folder of device subfolders. This is not the BUILDING_100 Overview-mirrored lab PDF. Figures use the PyPI Plotly helpers shared with the Railway UI.

AHU_1

Sensor checks

Checks for stuck, out-of-range, or stale sensors. Only failed checks are listed when something is wrong.

Fail. Outdoor air temperature: reading out of physical range (low 35.0, high 200.0).

Fail. Return air temperature: sensor stuck flat; Supply air temperature: sensor stuck flat.

Fail. Outdoor air temperature: reading jumped suddenly.

Fail. Outdoor air temperature: reading changed faster than expected.

Anomaly screening

A quick look at whether each varying trend is ordinary this month. Not an equipment fault list.

Mixed air temperature: looks normal.

Outdoor air temperature: looks normal.

Executive summary

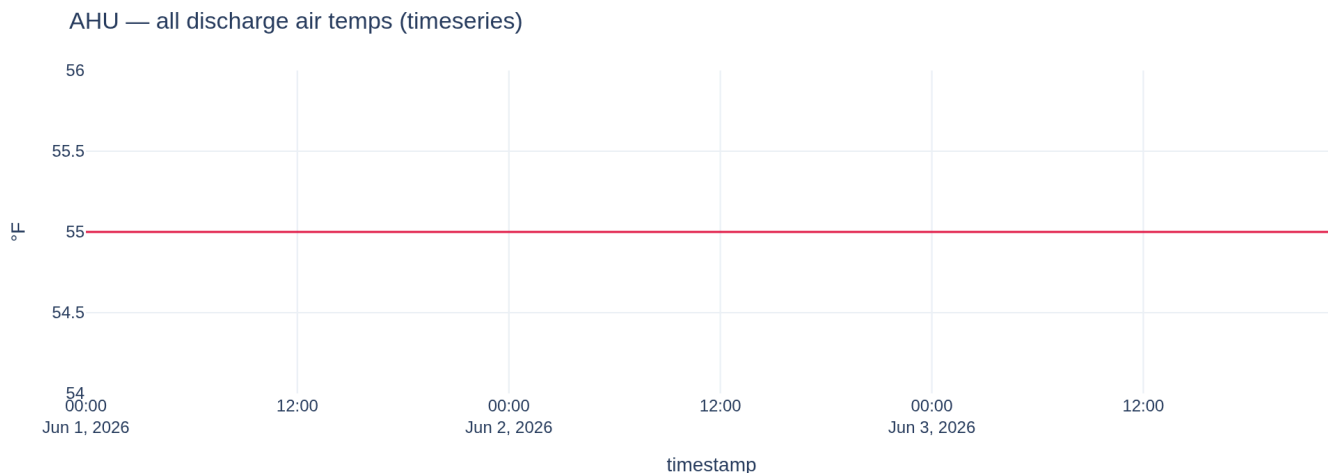
AHU_1 in June 2026 (72 samples, 72.0 h). Sensor checks found issues. Fail. Outdoor air temperature: reading out of physical range (low 35.0, high 200.0). Fail. Return air temperature: sensor stuck flat; Supply air temperature: sensor stuck flat. Fail. Outdoor air temperature: reading jumped suddenly. Fail. Outdoor air temperature: reading changed faster than expected. Anomaly screening: screened trends look normal. These screening notes are not equipment faults. Confirmed operating findings: Duct static below SP at full fan (GL36 A) 72.0 h; Economizer stuck closed 26.0 h; Always-on fan or pump runtime 72.0 h. The fan was on for 100.0 percent of the samples.

RCx week (2026-06-01 to 2026-06-03 UTC)

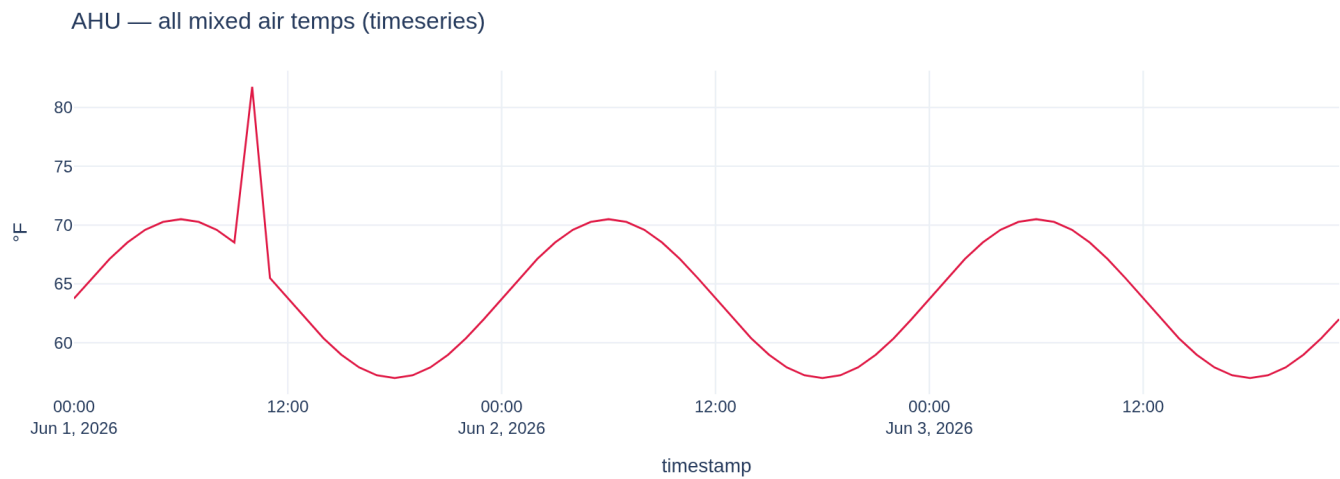
One representative week inside the filtered month, preferring fan-ON coverage. Lines use the PyPI Plotly helpers shared with Railway RCx plots.

AHU — all discharge air temps (timeseries)

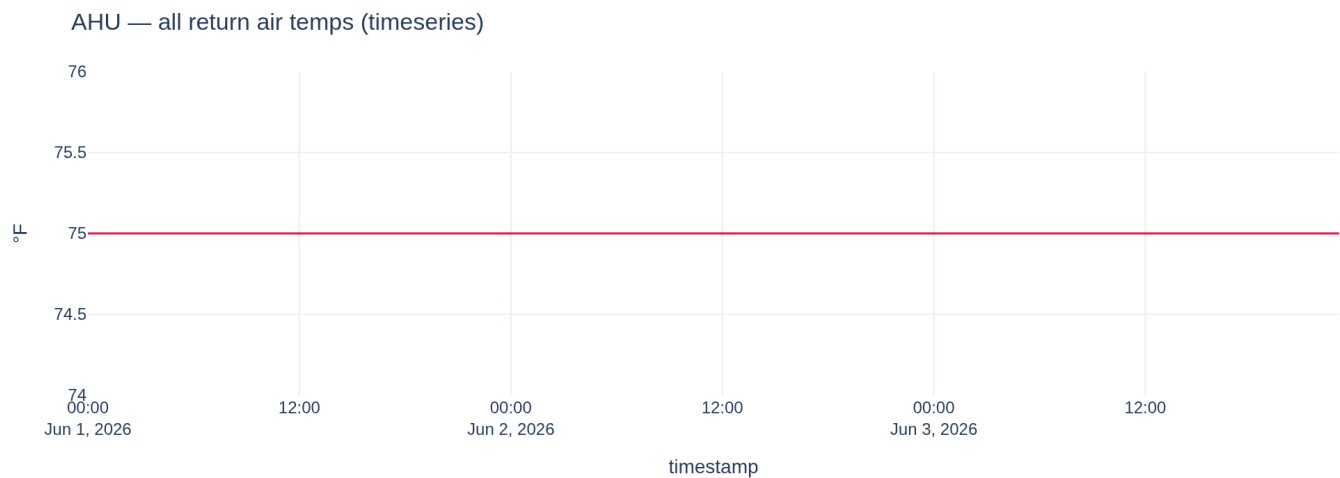
SAT / DAT for every AHU.



AHU – all mixed air temps (timeseries)
MAT across AHUs.

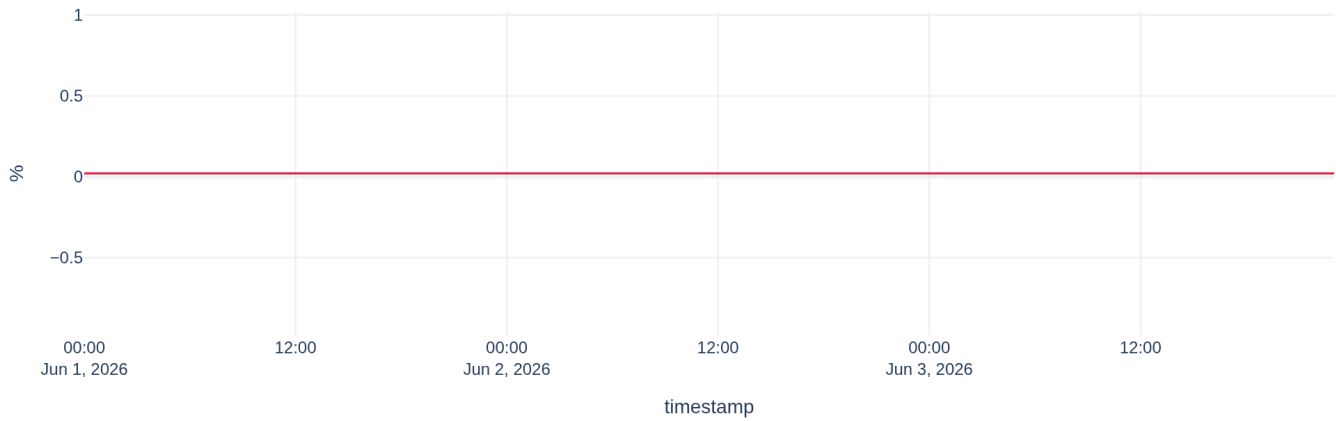


AHU – all return air temps (timeseries)
RAT across AHUs.



AHU – all OA dampers (timeseries)
OA damper % across AHUs (preferred % companion for SAT/MAT/RAT views).

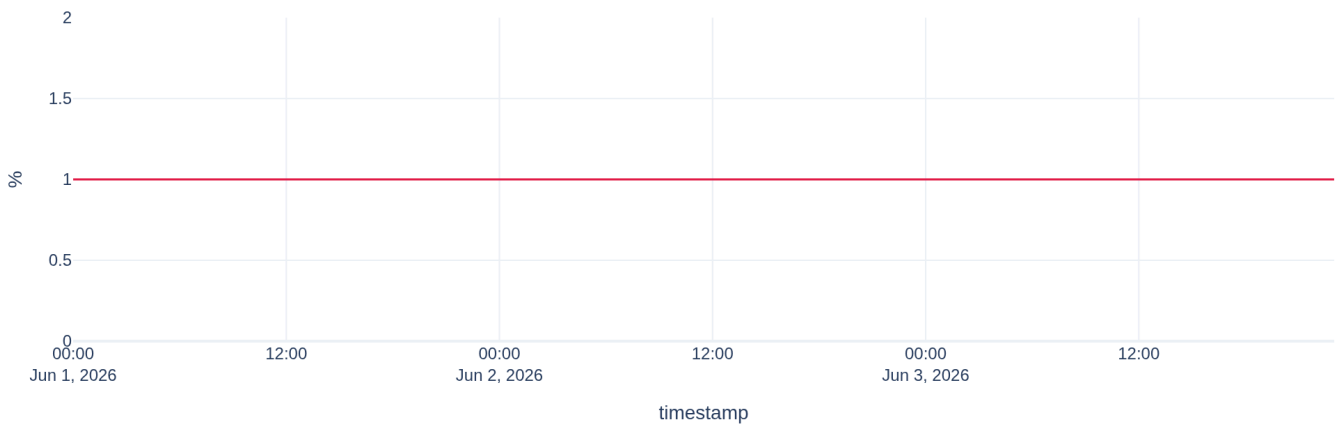
AHU — all OA dampers (timeseries)



AHU — all fan speeds (timeseries)

Fan command % across AHUs – pair with duct-static / FC1 pressure views, not SAT/MAT.

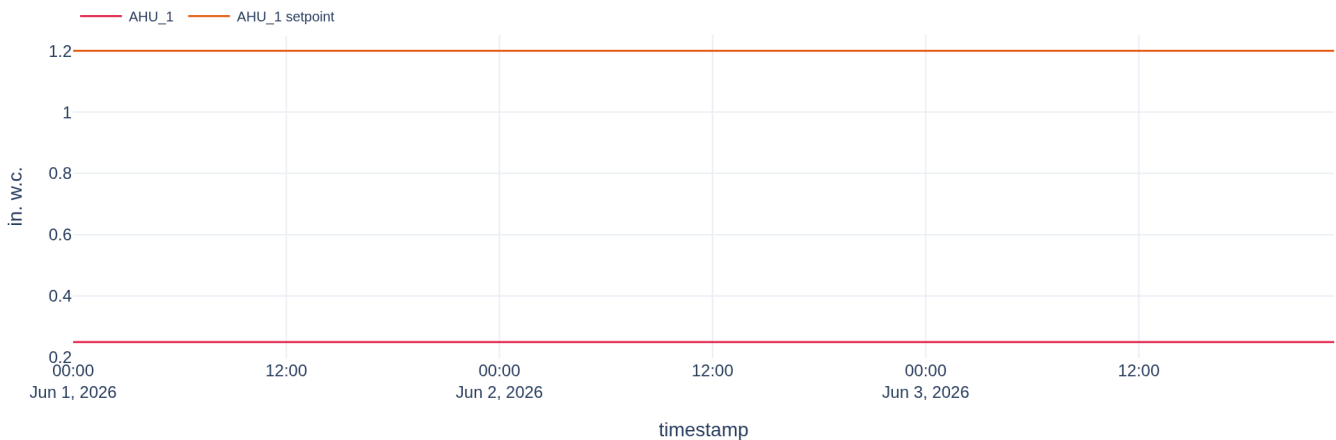
AHU — all fan speeds (timeseries)



AHU — duct static + setpoint (timeseries)

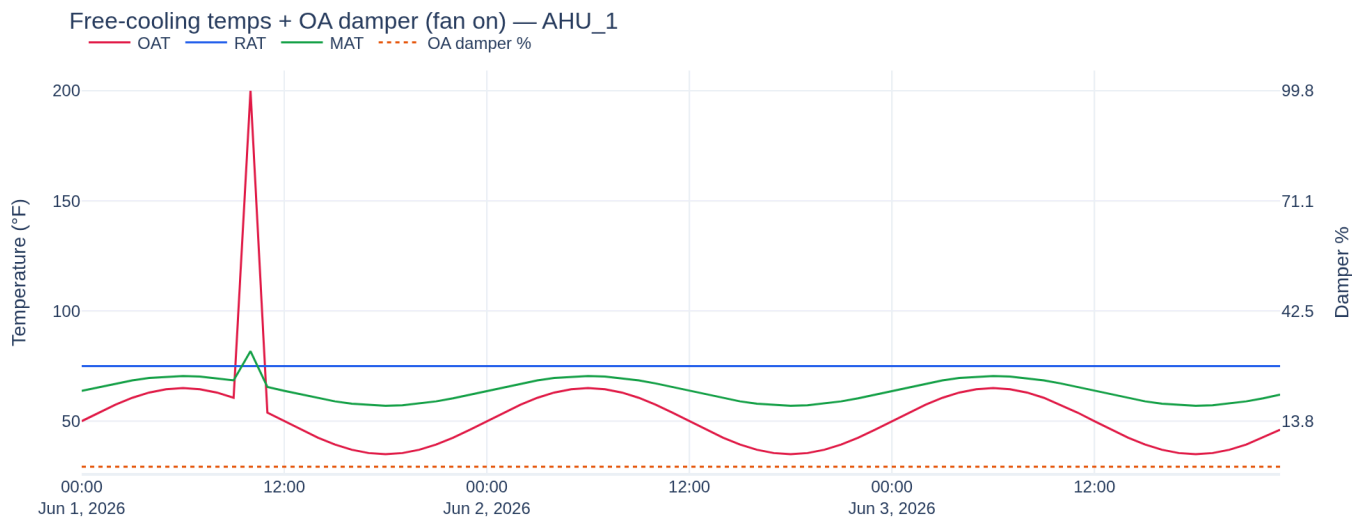
Duct static pressure over time per AHU, with the mapped setpoint as a companion trace – flat setpoint + high static while fan runs means a reset opportunity.

AHU — duct static + setpoint (timeseries)



Free-cooling temps + OA damper

OAT, RAT, MAT, and SAT on the temperature axis; OA damper percent on the right axis.



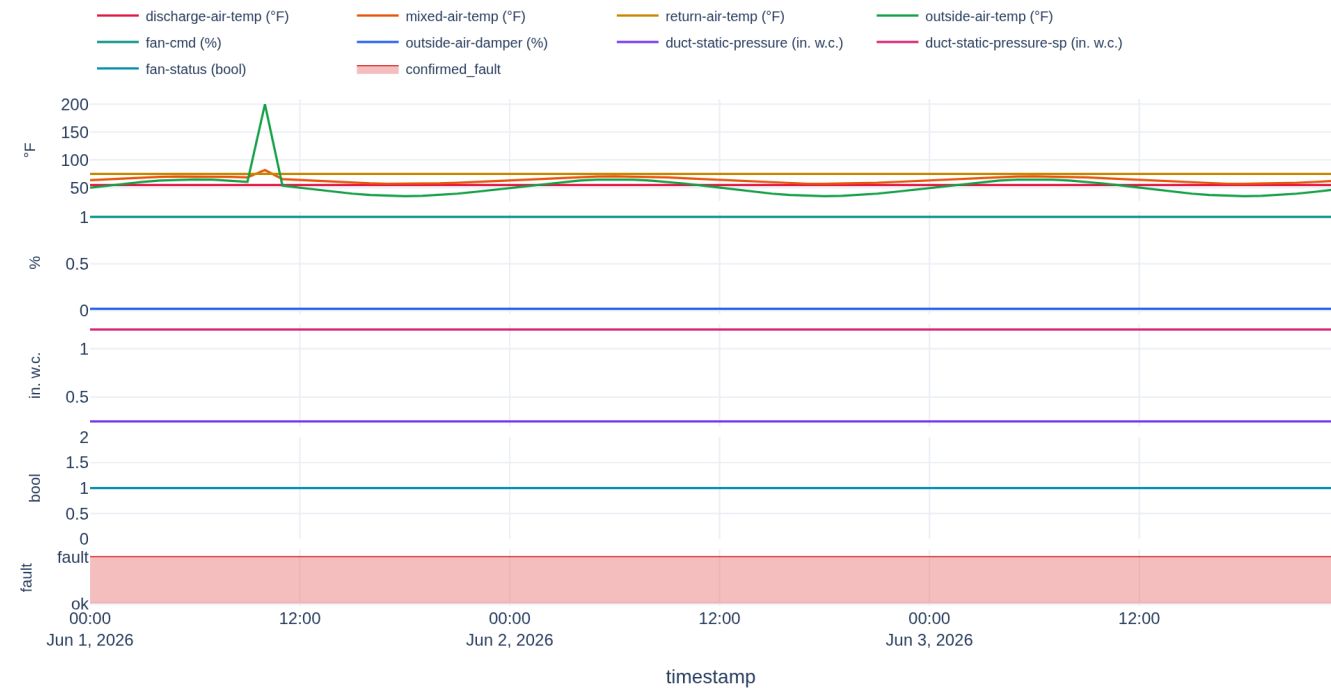
Confirmed faults

A figure is included only when the cookbook rule has confirmed fault hours in the filtered month.

FC1 – Duct static below SP at full fan (GL36 A)

Rule: Flags duct static too low while the supply fan is near full speed.

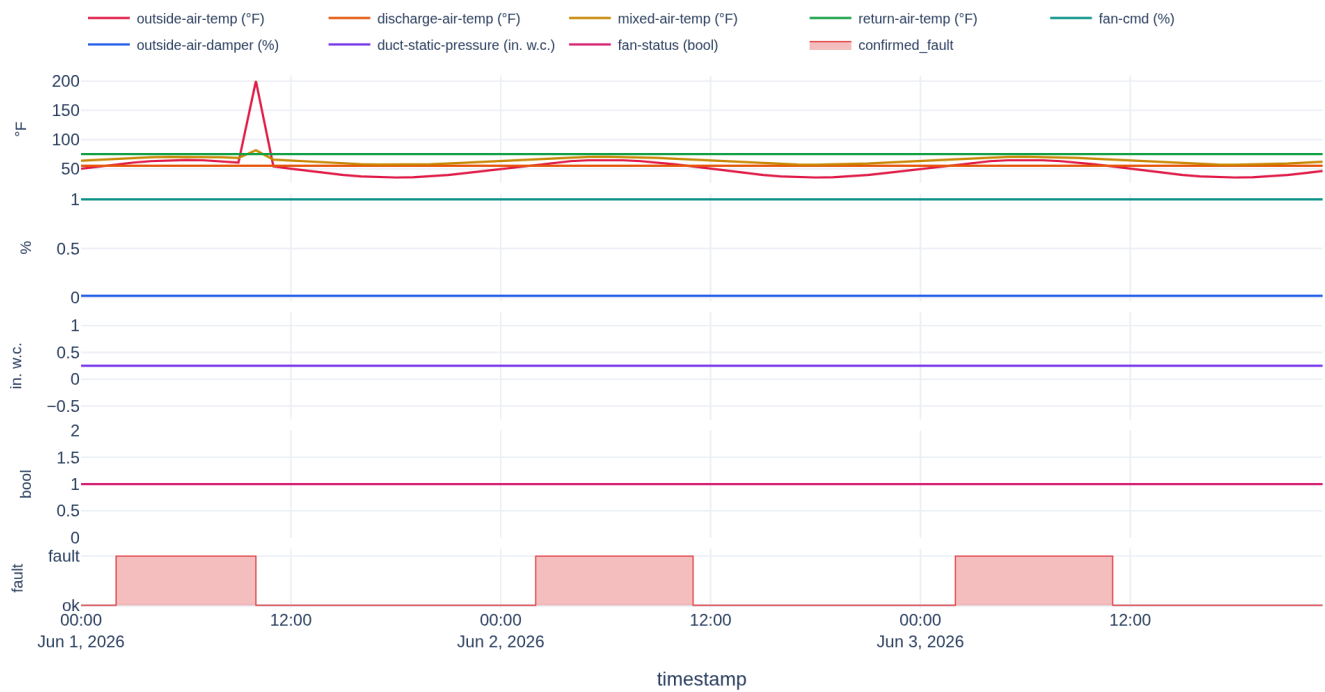
Data: 72.0 fault hours in 2026-06. 72.0h fault (100.0% of active)



ECON-1 – Economizer stuck closed

Rule: Fan on, OA damper < 5%, OAT > 55°F (should be economizing).

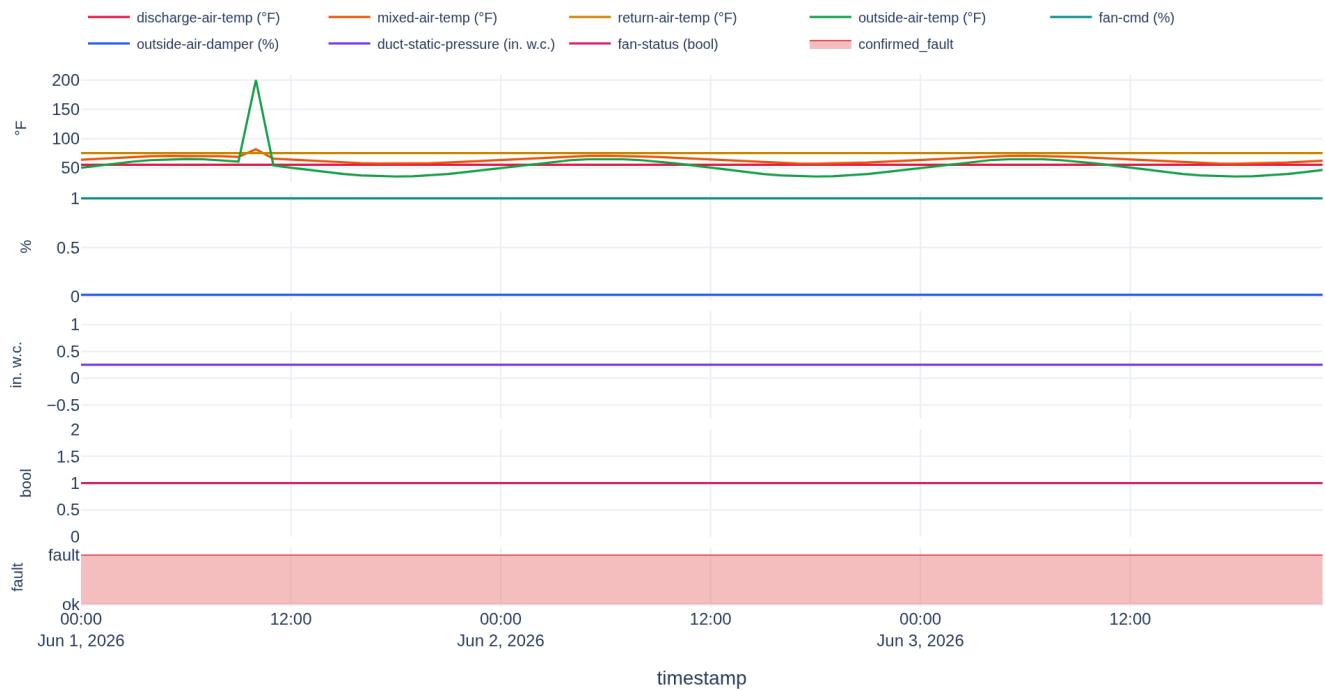
Data: 26.0 fault hours in 2026-06. 26.0h fault (36.1% of active)



SCHED-247 – Always-on fan or pump runtime

Rule: Flags fans or pumps that stay on for nearly the entire analysis window (24/7 runtime).

Data: 72.0 fault hours in 2026-06. 72.0h fault (100.0% of active)



Economizer delta scatter

economizer_delta_scatter: $x = \text{OAT} - \text{RAT} (\text{delta_or_f})$, $y = \text{MAT} - \text{RAT} (\text{delta_mr_f})$. Reference lines $y = \text{OA fraction} * x$ for 0/25/50/75/100% OA. Fan ON and $|\text{OAT}-\text{RAT}| \geq 10$ F. Viewport is the

bottom-left mixing quadrant only (both deltas ≤ 0 : $OAT \leq RAT$ and $MAT \leq RAT$). Do not plot $OAT - MAT$ vs $RAT - MAT$.

